FLATNESS OVER A QUASI-FINITE POLYNOMIAL BASE EQUALS COHEN-MACAULAYNESS OF THE EXPECTED LOCAL DIMENSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement refers to the definition of relative dimension at a point; it is recorded as a navigation stub with kind definition, and the argument using it stays inside the counted span.

Editorial note. The counted proof also refers to same-chapter statements by label, recorded the same way; no \cite appears, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- The dimension inequality for a quasi-finite map. Invoked in the counted proof at lines 33759-33761. Bounds the dimension of the local ring by that of the polynomial base.
- Flatness of a Cohen-Macaulay module over a regular local ring. Invoked at the end of the counted proof (lines 33783-33785). Converts the Cohen-Macaulayness and dimension condition back into flatness.

Source reference. Original definition 00QD: sources/stacks-algebra/pinned-original-source.tex, source lines 31367–31367. Navigation pointer for \ref{definition-relative-dimension} used inside the counted span (line 33753); the argument that consumes it is retained in the span.

Source reference. Original lemma 00R4: sources/stacks-algebra/pinned-original-source.tex, source lines 32871–32871. Navigation pointer for \ref{lemma-CM-over-regular-flat} used inside the counted span (line 33783); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00P1: sources/stacks-algebra/pinned-original-source.tex, source lines 28439–28439. Navigation pointer for \ref{lemma-dimension-at-a-point-finite-type-field} used inside the counted span (line 33772); the argument that consumes it is retained in the span.

Source reference. Original lemma 00QF: sources/stacks-algebra/pinned-original-source.tex, source lines 31484–31484. Navigation pointer for \ref{lemma-dimension-inequality-quasi-finite} used inside the counted span (line 33761); the argument that consumes it is retained in the span.

Source reference. Original lemma 00R5: sources/stacks-algebra/pinned-original-source.tex, source lines 27706–27706. Navigation pointer for \ref{lemma-finite-flat-over-regular-CM} used inside the counted span (line 33768); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a finite type algebra over a field k. Let $\varphi : k[y_1, \ldots, y_d] \to S$ be a quasi-finite ring map. As subsets of* $\mathrm{Spec}(S)$ *we have*

$$\{\mathfrak{q} \mid S_{\mathfrak{q}} \text{ flat over } k[y_1, \ldots, y_d]\} = \{\mathfrak{q} \mid S_{\mathfrak{q}} \text{ CM and } \dim_{\mathfrak{q}}(S/k) = d\}$$

For notation see Definition 00QD.

Proof. Let $\mathfrak{q} \subset S$ be a prime. Denote $\mathfrak{p} = k[y_1, \ldots, y_d] \cap \mathfrak{q}$. Note that always $\dim(S_{\mathfrak{q}}) \leq \dim(k[y_1, \ldots, y_d]_{\mathfrak{p}})$ by Lemma 00QF for example. Moreover, the field extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is finite and hence $\mathrm{trdeg}_k(\kappa(\mathfrak{p})) = \mathrm{trdeg}_k(\kappa(\mathfrak{q}))$.

Let $\mathfrak{q}$ be an element of the left hand side. Then Lemma 00R5 applies and we conclude that $S_{\mathfrak{q}}$ is Cohen-Macaulay and $\dim(S_{\mathfrak{q}}) = \dim(k[y_1, \ldots, y_d]_{\mathfrak{p}})$. Combined with the equality of transcendence degrees above and Lemma 00P1 this implies that $\dim_{\mathfrak{q}}(S/k) = d$. Hence $\mathfrak{q}$ is an element of the right hand side.

Let $\mathfrak{q}$ be an element of the right hand side. By the equality of transcendence degrees above, the assumption that $\dim_{\mathfrak{q}}(S/k) = d$ and Lemma 00P1 we conclude that $\dim(S_{\mathfrak{q}}) = \dim(k[y_1, \ldots, y_d]_{\mathfrak{p}})$. Hence Lemma 00R4 applies and we see that $\mathfrak{q}$ is an element of the left hand side. □